RESEARCH CONTINUATION / 60-DAY PROPOSAL

From evidence to integration.

ARGUS Surface Support | November–December 2026

An independent quality-assurance effort for traceable, reproducible checks of surface-prediction labels against masked CT storage. The objective is a useful maintained contribution—not a claim to have identified ink, or a request for special status.

24 / 24
Prespecified sampled cells

13
Stored prediction chunks

19,932,389
Independently counted positive labels

Measured October finding. Across the 13 stored prediction chunks in a frozen local PHerc. 211 sample, 8,519,612 positive labels (42.74%) fell at indices without corresponding stored masked-CT Zarr chunk objects. The finding was independently re-counted. It is not a scroll-wide rate, proof of false papyrus, or ink detection. Physical registration remains unverified.

The nearest prior art, vesuvius-phantom-audit, already covers broad phantom auditing, voxel checks, and cleanup. ARGUS must establish a specific complementary advantage or contribute a smaller improvement upstream.

The critical path

A reproducible public release → a matched baseline benchmark → independent registration/mask-semantics audit → one actionable researcher decision → independent transfer test → maintainable stable release.

The rule for every gate

When evidence is insufficient, publish the uncertainty and preserve a limited storage-support only interpretation. A technical failure is never rewritten as a scientific success.

Document intent: accompany the October Progress Prize evidence with a realistic route to community utility. Future work and outside collaboration are proposed, not claimed.

CRITICAL PATH / NOVEMBER & DECEMBER

Milestones with exit gates

WINDOW	WORK	EXIT / DECISION
G0 · Oct 23–31	**Release integrity**	Public source / immutable commit; license + data-rights check; documented Windows and Linux reproduction. Do not claim released until public.
G1 · Nov 2–6	**Equivalent benchmark**	Same inputs and masks as the existing audit; report agreement, runtime, memory, network bytes and failure handling. Choose complementary feature or upstream PR.
G2 · Nov 9–13	**Registration audit**	Independently assess axes, origins, transforms, source-volume identity, masked fill semantics and landmarks. Label REGISTERED / NOT REGISTERED / UNRESOLVED.
G3 · Nov 16–20	**Downstream demonstration**	At least one repeatable triage example and consistent machine-readable export; show a researcher decision, not merely a large number.
G4 · Nov 23–30	**Transfer study**	Freeze a second region or scroll before inspecting results. Record positive, negative, absent and unavailable data. Publish the complete denominator.
G5 · Dec 1–11	**Reproduce & harden**	Versioned inputs/outputs; clean Windows and Linux tests; source-byte replay; public issue tracker; invite optional external reproduction.
G6 · Dec 14–18	**Ship or merge**	Technical note, benchmark, limitations, upstream issue/PR or standalone versioned release. If novelty fails, merge or retire duplication.
Dec 21–31	**Buffer & review**	Maintenance, documentation and feedback follow-up; no dependence on prize decision or outside reviewer availability.

Independent R12 research remains off the critical path. Frozen 55-plane +13…+15 analyses can resume only after native-coordinate and geometric validation. Gate 7F stays blocked; there is no verified writing.

ACCOUNTABILITY / COMMUNITY CONTRIBUTION

Earn trust through useful work.

Five acceptance conditions

Correctness. Count identical raw data by two implementations and lock the inputs by hash.

Honest geometry. Never confuse absent masked storage with physical absence or invalid papyrus.

Fair novelty. Benchmark against published tools; prefer a useful upstream patch over duplicate software.

Practical integration. Provide at least one machine-readable result that changes a real downstream review decision.

Transparent participation. Publish failures, unresolved registration, known limitations and reviewer feedback accurately.

Invitation for technical review

> “I am contributing ARGUS as a reproducible quality-assurance tool for surface predictions. I would especially welcome correction of coordinate assumptions, a suggestion for a useful integration point, or an independent attempt to reproduce one fixed case. I recognize the existing phantom-audit work and am happy to adapt, upstream, or retire overlapping features. My aim is to remain a constructive contributor beyond the prize decision.”

Small, concrete requests

(1) Authoritative prediction-to-CT transforms and mask semantics. (2) One existing workflow that would benefit from a source-auditable support report. (3) Optional review of a matched benchmark or small proposed upstream change. No private data, endorsement, allocation or appointment is requested.

What this proposal does not claim

It does not claim a public GitHub release, community adoption, external validation, proven phantom papyrus, or discovered ink. Those are separate evidence gates. Existing source and result links must be made public and verified before submission.

Sources: Vesuvius Challenge Progress Prize rules and FAQ; Schurkai vesuvius-phantom-audit; ARGUS 0.4.1 prespecified held-out and six-case reports. Full work packages, risks and gates: the accompanying Markdown proposal.